FEDERICO BRANCASI

Research Engineer at Microsoft

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EXPERIENCE

Research Engineer — February 2026 – Present
Microsoft

- Fine-tuning language models and optimizing prompts for **VS Code's AI features**, reducing Time-To-First-Token (TTFT) and improving response quality for inline code completion.
- Building **data pipelines for model fine-tuning**, including data collection, cleaning and evaluation workflows to enhance the developer experience.

Machine Learning Researcher — August 2025 – February 2026
CERN (European Organization for Nuclear Research)

- Developed **QuantDiff**, an automated mixed-precision quantization framework that finds the optimal bit allocation by balancing each layer's sensitivity against its computational cost, enabling precise model compression.
- Created **Neural Anomaly Metric**, a new evaluation metric that better captures how quantization affects model accuracy, addressing limitations of existing metrics.

Machine Learning Researcher — January 2025 – July 2025
ETH Zurich (Federal Institute of Technology Zurich)

- Built a custom backend for AMD's Brevitas (PyTorch Quantization Framework) to enable **automated neural network quantization and deployment on low-power embedded processors**.
- Conducted comprehensive benchmarking of different quantization schemes, with particular attention to non-linear operator quantization and automatic network optimization for embedded deployments.

Research Assistant — February 2023 – July 2024
University of Trento

- Modernized Computational Logic course materials and led the development of a **new textbook**, integrating Knowledge Graph methodologies and creating advanced visualizations with LaTeX, resulting in a **30% increase in student pass rates**.

EDUCATION

M.Sc. in Computer Science — September 2023 – January 2026
ETH Zurich, University of Trento, Aalto University of Helsinki, Eötvös Loránd University of Budapest

- Selected as one of the **200 Students in Europe** for this program co-founded with the european union, studying two years across two different cities — in my case, Trento and Budapest.
- Key role as a **Student Ambassador**: Promoted program benefits to engage prospective students.
- Master Thesis on ML Quantization developed at **ETH Zurich**. Summer school on Smart Cities at **Aalto University**.
- **Main Courses**: Deep Learning, Advanced Deep Learning, Machine Learning, Statistical Learning, Statistical Methods, Cyber Security Risk Assessment, Distributed Systems, Business Intelligence, Cloud Systems, DevOps Engineering.
- *Additional: Postgraduate Diploma at Trento School of Innovation — evening program focused on soft skills led by industry professionals. Courses: Data Governance, Emotional Intelligence, Collective Creativity in Organizations.*

B.Sc. in Computer Science — September 2020 – September 2023
University of Trento

- **Main Courses**: Machine Learning, Operating Systems, Data Structures and Algorithms Design, Object-Oriented Design, Networking, Software Engineering, Databases Systems, Formal Languages and Compilers, Computer Architectures.

EXTRACURRICULAR

Anthropic's Claude 4 Hackathon London Winner [link]
Selected as winner at Anthropic's inaugural student hackathon at the London School of Economics and Political Science (LSE), building an intelligent security system with natural language control. Developed a full-stack application combining real-time video monitoring with Claude's conversational AI capabilities in just an hour.

AWS MLOps (Machine Learning Operations) Engineer Path [link]
Obtained **AWS Certified Cloud Practitioner**, **AWS Certified AI Practitioner** and **AWS Certified Solutions Architect**, currently pursuing additional AWS certifications related to Machine Learning and Cloud Computing.

AI Hackathon Munich Winner (Organized by AWS, OpenAI, Lovable, n8n, ElevenLabs, TUM.ai) [link]
Won the prototyping sprint at CDTM Munich, creating a complete application that transforms any data source (YouTube videos, websites, PDFs, CSVs, etc.) into professional business reports and presentations with one-click AI automation in just 6 hours, demonstrating end-to-end product development from concept to working prototype.

2nd Place at Startup Lab Competition [link]
Secured 30K euros pre-incubation funding for an innovative agritech solution, developing an autonomous harvesting prototype that combines YOLO object detection, ROS, and SLAM for precision agriculture.

Industrial AI Challenge Winner [link]
Led the development of an award-winning AI solution for BLM Group's inventory management, implementing machine learning models that reduced waste by 60 percent, improving overall supply chain efficiency.

SKILLS

Python | C/C++ | JavaScript | TypeScript | SQL | LaTeX | Bash | PyTorch | TensorFlow | LLMs | Transformers | Vision Transformers | ResNet/CNNs | Diffusion Models | Quantization | Fine-tuning | Prompt Engineering | RAG | Generative AI | NLP | Scikit-learn | Computer Vision | Model Compression | AI Agents | Anthropic Claude API | OpenAI API | Ollama | FastAPI | React | Node.js | Flask | n8n | LangChain | HuggingFace | Streamlit | Gradio | Tailwind CSS | Docker | Git | AWS | MLOps | CI/CD | GitHub Actions | Pandas | NumPy | Seaborn | Jupyter | Matplotlib | Plotly | VS Code | Linux | TensorBoard | pytest | Cloud Computing | Containerization | Vercel | Vector Databases | Supabase | REST APIs | WebSockets | MongoDB | PostgreSQL | ONNX | Technical Writing

SELECTED PROJECTS

DeepQuant Neural Network Quantization Library [link]

- Led the development a comprehensive Python library for converting quantized models with AMD's Brevitas library to true integer-only representations, enabling efficient deployment on embedded systems without floating-point units.
- Implemented advanced graph transformation algorithms including quantization node splitting, dequantization unification, and custom forward pass injection for neural network layers, supporting complex architectures like CNNs such as ResNet and Transformer-based models including Vision Transformers.

Neural Anomaly Metric [link]

- Developed a new evaluation metric that better measures how quantization affects model accuracy, using anomaly detection to identify quality degradation that existing metrics fail to capture.
- Implemented support for multiple neural network backbones (Vision Transformers, ResNets, EfficientNet), enabling per-image quality assessment with statistical validation across different compression levels.

QuantDiff Mixed-Precision Framework [link]

- Built an automated mixed-precision quantization framework that optimally allocates bits across neural network layers by balancing each layer's sensitivity against its computational cost.
- Enabled precise model compression to any target size budget, supporting 4, 8, and 16-bit precision levels with quality assessment using FID and CLIP metrics.

Deep Learning Study Book [link]

- Authored a comprehensive deep learning textbook using LaTeX and TikZ, creating detailed technical illustrations of neural architectures and advanced ML concepts, publicly available in the University of Trento's shared resources.
- Developed extensive documentation of deep learning principles, including custom visualizations of transformers, CNN architectures and optimization techniques, serving as a key learning resource for future students.

Enhanced File Explorer for Chrome [link]

- Developed a browser extension to enhance local file browsing experience across Chromium browsers.
- Gathered more than 1000 daily users and surpassing 5000 installations on Chrome Web Store.